Hydrogen and Power-to-X

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JOJEN

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Introduction

Hydrogen energy

Hydrogen has been envisioned as an energy carrier long before the present debate on the climate and the green transition. Henry Cavendish[1] was the first to recognize hydrogen gas as a distinct substance in 1766, and Antoine-Laurent de Lavoisier[2] gave hydrogen its name in 1788. The first example of hydrogen by electrical energy dates back to 1789 when van Troostwijk and Deiman[3] used an electrostatic generator to provide electricity to a simple electrolyzer with gold electrodes. This was shortly followed by Nicholson and Carlisle[4] in 1800 with the help of the volta pile, the first battery and the first stable current source discovered the same year. The first primitive fuel cell, as an electrolyzer cell in reverse mode, was presented somewhat later in 1839 by Christian Friedrich Schönbein and Sir William Robert Grove[5].

The reaction of a hydrogen powered fuel cel is

$$2H_2 + O_2 \rightleftarrows 2H_2O + \text{energy} \quad (1)$$

And the reaction of a water electrolyzer is the opposite

$$2H_2O + \text{energy} \rightleftarrows 2H_2 + O_2 \quad (2)$$

Interestingly, already in 1874, the French author Jules Verne mentioned hydrogen as a future fuel in "The mysterious island". Practical electrolyzers were developed in the late years of the 19th century and scaled up in the early 20th century, led by Norsk Hydro in Norway. The discovery of the Haber–Bosch process for ammonia production directly from hydrogen and nitrogen in 1909 lead to an increasing demand for hydrogen. Today ammonia is the second-largest bulk industrial chemicals produced worldwide (by mass, and only second to sulfuric acid), but the hydrogen source is fossil via reforming for cost reasons. The main use for ammonia is for production of fertilizers.

The idea of hydrogen as an energy carrier was put is practice during the space missions in the 1960s. Liquid hydrogen for propulsion via rocket engines and for onboard electricity via early fuel cells. Later, in the 1970s during the oil crises, the idea of the hydrogen society, where hydrogen made from renewable energy was seen as a possible clean and abundant alternative to oil, saw the first hype.

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[1] Henry Cavendish (1731-1810). British scientist. Made hydrogen from dissolution of zinc in hydrochloric acid: $Zn + 2HCl \rightarrow H_2 + ZnCl_2$. He subsequently burned it and formed water.

[2] Antoine-Laurent de Lavoisier (1743-1794) French scientist. Gave 1788 hydrogen its name, "hydrogène" from Greek "*hydro*" (Water) and "*genes*" (to form or borne from). Lavoisier came to a tragic end during the French revolution.

[3] Adriaan Paets van Troostwijk (1752-1837) Dutch merchant/chemist. Jan Rudolph Deiman (1743-1808) Dutch/German doctor/chemist.

[4] The British scientists William Nicholson (1753-1815) and Anthony Carlisle (1768-1840) are often credited the inventors of the water electrolyzer although they built on the discoveries of van Troostwijk and Deiman.

[5] German chemist Christian Friedrich Schönbein (1799-1868) and English lawyer Sir William Robert Grove (1811-1896). Grove called his fuel cell a gas battery.

The principle for hydrogen is extremely simple. Water is split electrochemically into its constituents, hydrogen and oxygen and the electrical energy consumed in the process is stored as chemical energy in the produced hydrogen. When needed, the process can be reversed, and the electrical energy reclaimed my means of a fuel cell. See Figure 1. The produces oxygen can be vented to the air and later supplied by the air when needed as it contains around 20% oxygen.

Of course, the description of the hydrogen cycle is simplified, not form a chemical point of view, but from a practical and energetic point of view. If a process shall proceed at a practical rate, some entropy will be produced, and reality, we will see that unfolding as waste heat. Each step in the process is thus accompanied by energy losses and the electrical energy returned after one such cycle is less than half of the primary electrical energy. It can even be much less, depending on the cell technology and the way hydrogen is stored. If hydrogen is not stored, there is no need to produce it. The reasons behind the losses are plenty, but significant contributions come from electrode overvoltages and ohmic voltage loss in the electrolyte.

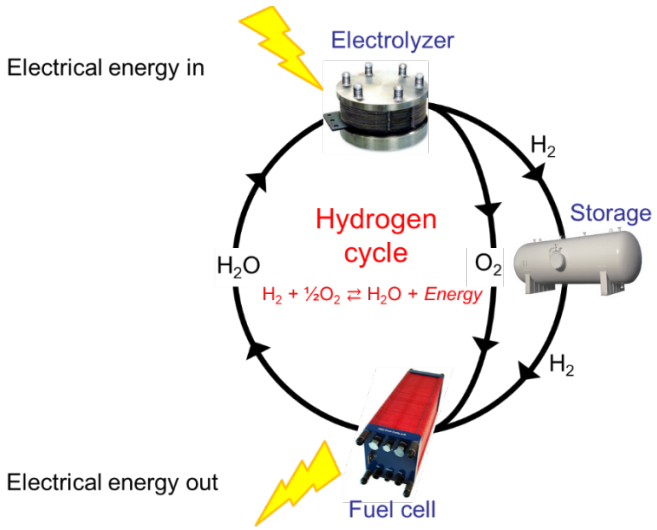


Figure 1. The hydrogen energy cycle. Production from water and electrical energy in an electrolyzer, storage in a tank and conversion back to water and electricity in a fuel cell.

Implementation of hydrogen as an energy carrier

The beauty of the hydrogen cycle is that the starting material is plain water, which is highly abundant and the process does not involve any harmful substances. However, there are several factors delaying the implementation of hydrogen at scale in the energy system.

1. Despite renewable energy from solar and wind has become cost competitive; the cost of electrolytic hydrogen is still significantly higher than that of fossil-based hydrogen.
2. Fuel cells and electrolyzer are not yet produced in large series and therefore they are still expensive.

3. Rolling out hydrogen as an energy carrier requires huge infrastructure investments. This is often referred to as the chicken-and-egg problem. Who will adapt to hydrogen without the infrastructure, who will build the infrastructure when there is no hydrogen and who will produce hydrogen if there are no of-takers.

Years back the anticipation was that hydrogen vehicles would make the be the first hydrogen breakthrough, but today the fuel cell vehicles are losing the competition to battery electric vehicles - at least on the commuter market. Heavy duty applications may be another story like other applications where the energy density of batteries is not sufficient. A hydrogen tank has higher energy density than the most advanced batteries and it is much cheaper.

Transportation of energy over long distances is more energy efficient than high voltage cables and in addition, the pipeline is cheaper per power capacity. Io this background a possible early application of hydrogen at scale is for industrial use in e.g., refineries, steel works and ammonia plants.

Power-to-X

Power-to-X (*PtX, P2X*) is a common term for all the different processes of converting electrical power into matter. Previous more specific terms were *power-to-gas*, *power-to-liquid*, *power to fuel*, *power-to-chemicals* etc. *Electrofuels* and *e-fuels* are fuels made by power-to-X. X can be just hydrogen and it can be ammonia, methanol, methane or even synthetic gasoline or diesel. The fuels or chemical other than hydrogen will normally require one or more subsequent synthesis steps involving a carbon source (possibly CO_2) or nitrogen from the air (Figure 2). As the direct use of hydrogen may be less widespread than originally anticipated, at least in the transport sector, focus is partly changing to derived molecules for shipping, aviation and industry. The demand in these sectors is immense, and the declining interest for hydrogen in some other sectors is often misread as if hydrogen and electrolysis in not relevant anymore as part of the solution in the green transition. The main pathways in the power-to-X concept are sketched in Figure 3.

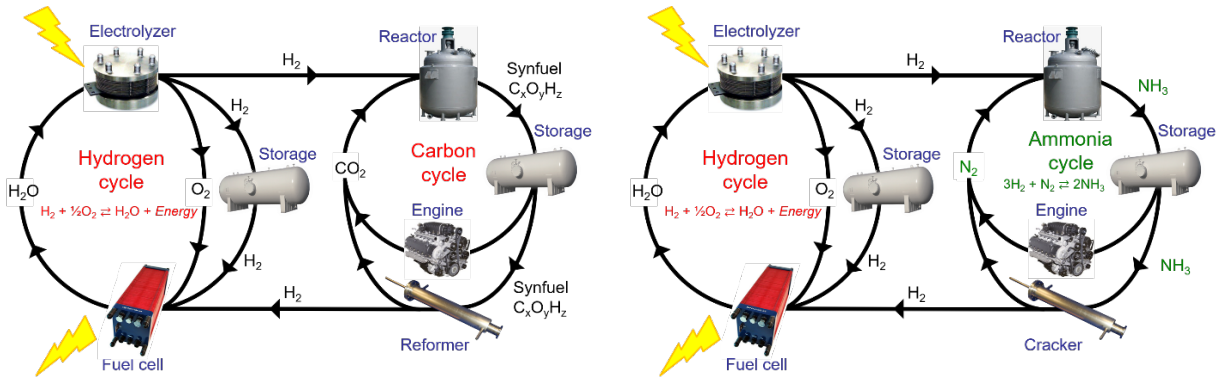


Figure 2. The hydrogen cycle from Figure 1 elaborated with a carbon cycle (left) and a nitrogen cycle (right).

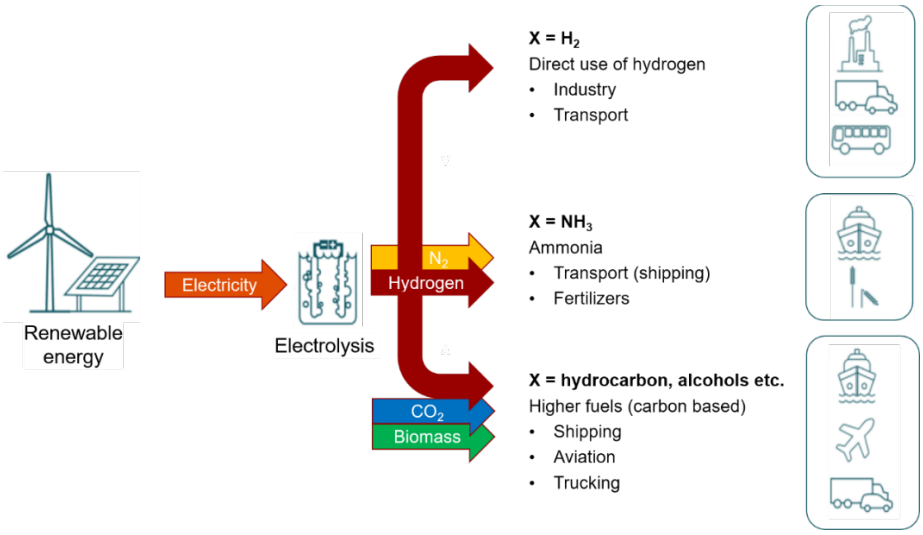


Figure 3. The main pathways in the power-to-X concept. Carbon based chemicals may also be used as feedstock in the chemical industry.

Table 1. The colours of hydrogen (*from Weisenburger 2025, DOI: 10.1016/j.erss.2025.104346,* CC BY 4.0)

	Color	Definition
Overall common understanding	green	hydrogen directly or indirectly produced using solar, wind and water power without generating direct CO2 emissions or other waste
	grey	hydrogen from steam reforming of natural gas without the prevention or containment of CO2 emissions
	blue	hydrogen from steam reforming of natural gas with additional CCS to prevent CO2 emissions (regardless of the capture rate)
	turquoise	hydrogen from pyrolysis of natural gas with generation of solid carbon as waste product
Coal-based hydrogen	brown	hydrogen from gasification of lignite
	black	hydrogen from gasification of black coal
Nuclear power based hydrogen	pink	hydrogen from electrolysis using electrical energy from nuclear power plants
	purple	hydrogen from combined chemo-thermal electrolysis based on nuclear power
	red	hydrogen from high temperature catalytic splitting of water based on nuclear power
Other introduced colors	yellow	hydrogen from electrolysis using a mixed carbon electrical grid
	white	naturally occurring hydrogen
	orange	hydrogen from conversion of biomass

The colours of hydrogen

Hydrogen is assigned a colour depending on its origin. The most common shades are green, blue and grey hydrogen referring to hydrogen produced by electrolysis via renewable electricity, by natural gas reforming with carbon sequestration, by natural gas reforming without carbon sequestration, respectively. Numerous colour schemes have been published with some differences regarding the less-common types of hydrogen. Table 1 sums up the general agreement based on a comprehensive literature study.

Fuel cells in brief

A fuel cell is a device that converts chemical energy of a fuel directly into electrical energy by an electrochemical process. In conventional processes for generation of electricity from a fuel, the fuel is combusted, and the heat liberated is converted to mechanical energy in a heat engine (a turbine or a piston engine). The mechanical energy is converted further into electricity by means of a generator. A fuel cell can extract the chemical energy or part of it directly while converting the fuel to its reaction products. These reaction products are in most cases, although not all, the same as the combustion products. The two pathways are visualized in Figure 4.

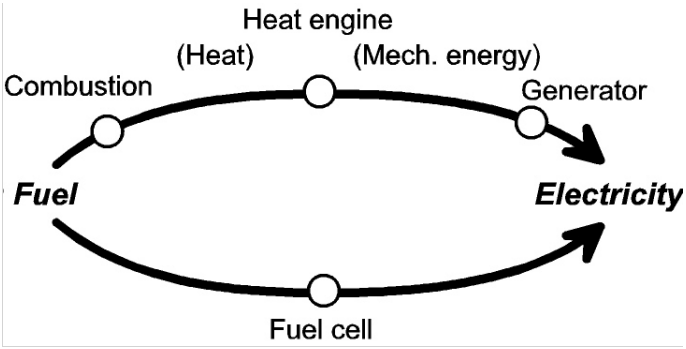


Figure 4. The traditional pathway from chemical energy to electrical energy and the direct conversion pathway in a fuel cell.

Since conversion in a fuel cell does not happen through a state of thermal energy (heat), the Carnot limitation does not apply to a fuel cell and the ultimate thermodynamic conversion efficiency - fuel to electricity - is only limited by the Gibbs free energy change of the equivalent combustion process.

Fuel cells have fascinated scientists, engineers and other people with an interest in technology for decades since their early applications as power sources in the space programmes. The main advantages are

- The possibility for higher electrical efficiency than with thermal engines
- Full scalability from below 1 W to kW or even MW with practically the same performance characteristics. In contrast, combustion engines gain efficiency with size, mostly due to lower relative heat loss (The huge shipping engines are actually very efficient)

- Silent operation and no vibrations with “no moving parts” (not quite true when the system around with blowers is considered)
- The perfect match with hydrogen as fuel with emission of only pure water - the clean hydrogen energy concept.

Fuel cells can in principle be constructed to convert any fuel, or any energy releasing chemical process in which electrons are exchanged between species, but most commonly they are powered by hydrogen.

Electrolyzers in brief

Electrolyzers are devices that produce a product by forcing a chemical process by electrical power. They are like fuel cells but optimized for production of specific molecules instead of electrical current. They are not necessarily just the opposite of fuel cells but find use is several branches of the chemical industry. Table 2. Water electrolysis and selected important electrolysis processes in the industry. In addition to the well-established technologies in the table. Electrolysis of CO_2 into valuable chemicals is under development but not yet implemented industrially. In this context, we will only address water electrolysis.

Table 2. Water electrolysis and selected important electrolysis processes in the industry. Main desired products in bold.

Electrolysis process	**Overall reaction**	**Oxidation product**	**Reduction product**
Water electrolysis	$2H_2O \rightarrow 2H_2 + O_2$	O_2	**H_2**
Chlor-alkali process	$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$	**Cl_2**	H_2
Aluminium production (the Hall-Heroult process)	$2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$	CO_2	**Al**
Magnesium production	$MgCl_2 + \rightarrow Mg + Cl_2$	Cl_2	**Mg**
Sodium production	$2NaCl \rightarrow 2Na + Cl_2$	Cl_2	**Na**

The advantages of water electrolysis are

- Electrolyzers offer means for storing renewable electrical energy as hydrogen, and it is generated form an abundant low-cost material, water.
- The industrial demand for hydrogen is vast, and electrolysis allows for CO_2-free hydrogen production.
- Like fuel cells, they can be manufactures at very different scales ranging from kW to MW.
- The modular structure makes then suitable for very large hydrogen production plants.
- In the green transition of the energy system, electrolysis is the only realistic option for the conversion of renewable electricity to fuels and chemicals at scale.

The function of the cells

The single cell

The core of the conversion cell is the single cell constituting the electrodes and the electrolyte that separates them. See the schematic of a fuel cell in Figure 5. In chemical processes where oxidation and reduction take place (redox processes), these are always coupled to ensure charge balance in the end. In conversion cells oxidation and reduction are taking place at separate electrodes and electrons exchanged via the external electrical circuit. The charge balance is secured by exchange of charge by ions through the electrolyte. The three most common processes for forming or splitting water are listed in Table 3. Fuel cell mode involves the reactions from left to right and electrolyzer mode, right to left. Note that in the three examples, the electrode processes are varying, but that the total process is the same in all cases. Note that H^+ and O^{2-} are not both formed as free ions in the same process. In the example with an acidic cell in Figure 5, O^{2-} is formed as an integral part of the water molecule. The most common cell types are shown schematically in Figure 6 for fuel cells and Figure 7 for electrolyzers. In addition to these, there are three more fuel cell types, the alkaline fuel cell, which is basically the reverse of the alkaline electrolyzer, the phosphoric acid fuel cell, which is like the proton exchange membrane fuel cell but with liquid phosphoric acid as electrolyte, and the molten carbonate fuel cell with a more complicated reaction scheme involving CO_2.

Table 3. Reaction schemes for fuel cells and electrolyzer cell with different ion conducting electrolytes. For fuel cells the reactions process from left to right and for electrolyzers from right to left.

Proton conducting electrolyte (acidic). Ion conducted: H^+	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 \rightleftarrows 4H^+ + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$4H^+ + O_2 + 4e^- \rightleftarrows 2H_2O$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$
Oxide ion conducting electrolyte. Ion conducted: O^{2-}	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 + 2O^{2-} \rightleftarrows H_2O + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$O_2 + 4e^- \rightleftarrows 2O^{2-}$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$
Hydroxide ion conducting electrolyte (alkaline). Ion conducted: OH^-	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 + 4OH^- \rightleftarrows 4H_2O + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$O_2 + 2H_2O + 4e^- \rightleftarrows 4OH^-$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$

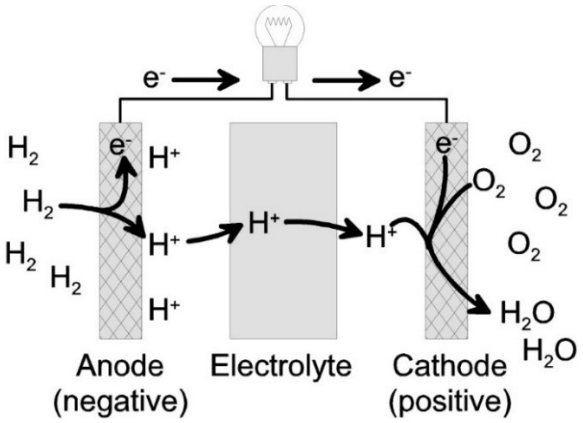


Figure 5. Schematic of a fuel cell with a proton conducting electrolyte. In an electrolyzer the processes and the arrows are opposite, and the lightbulb is replaced by a current source. With the opposite process the labels anode and cathode swap too.

Electrolytes

Electrolytes can be liquids contained in an inert porous matrix or they can be solids. They conduct specific ions. Liquid electrolytes are aqueous solutions of acid or base or in some cases a pure liquid acid. Acidic electrolytes conduct protons and alkaline (i.e., basic) electrolytes typically hydroxide ions, OH^-. The solid part of the electrolyte system (the porous matric or the entire ion-conducting material) moreover serves as a separator preventing the electrodes form short circuiting and the involved gasses from mixing. Historical alkaline electrolyzers used a cloth of asbestos soaked in aqueous potassium hydroxide as separator, but today these are replaced by porous polymeric membranes called *diaphragms*. The most common solid electrolytes are the proton exchange membrane, a specialty polymer which conducts protons, and the solid oxide electrolyte which is a ceramic oxide ion conductor. The latter builds on the mobility of oxide ions in the metal oxide crystal lattices at high temperature. In Table 3 with the reaction schemes for the cell reactions it can also be seen which ions are conducted.

Electrodes

Electrodes are often complicated structures. The electrochemical processes are carried out close to the electrolyte and are always promoted by catalysts at the surface of the electrode. Catalyst can be discrete catalytic particles or active surface layers applies at the part of the electrode in contact with the electrolyte. Th catalyst must be in good electrical contact with the electrode and accessible to all reactants of the reaction (Table 3), i.e., water, gasses and ions. Consequently, the electrode hosts not only the electrochemical processes, but also multiple transport processes as seen in Figure 8. Reactants and products must be transported in and out of the porous structure through the pores and ions must be transported between the catalyst layer and the bulk electrolyte. Moreover, electrons and heat must be conducted through the solid parts.

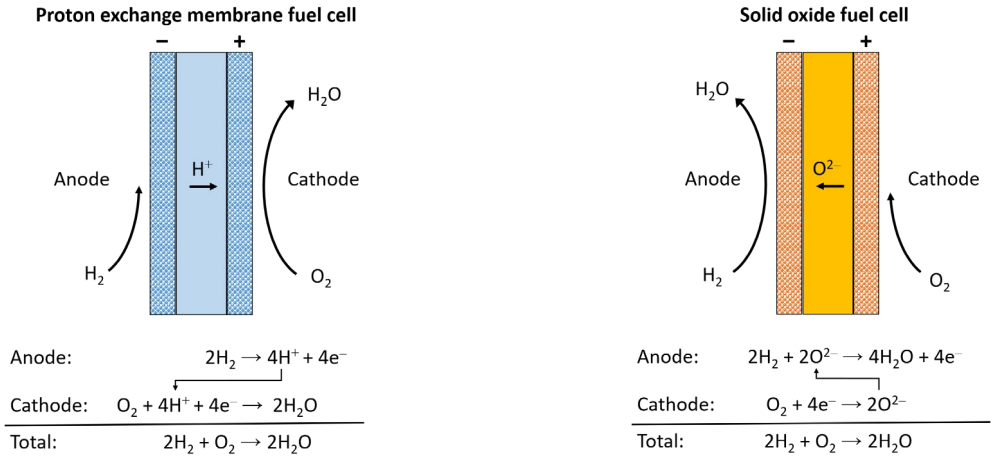


Figure 6. The two most important fuel cell types. Both types apply porous electrodes and non-porous ion-conducting membranes.

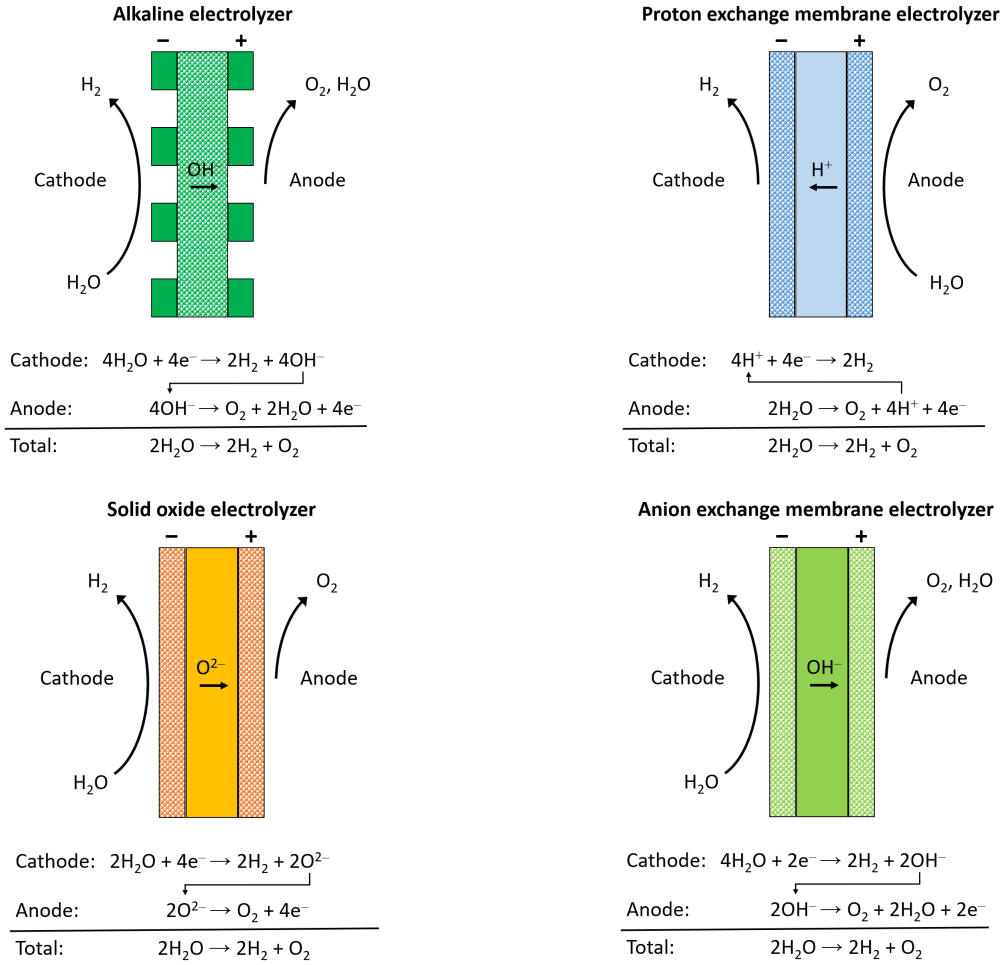


Figure 7. The four most important electrolyzer types. The alkaline electrolyser has a porous electrolyte filled separator (a diaphragm) and electrode of perforated plated or mesh, the other three cells have porous electrodes and non-porous ion-conducting membranes.

In fuel cells and steam-fed electrolysers, the catalysts must be connected to an electron conducting phase, an ion conducting phase and a gas phase. The places where that is fulfilled is called the *triple phase boundary*.[6] This is an important design element. In electrolyzers operated on liquid water, the three phase are coexisting as well, but we don't necessarily have the triple-phase boundary because the electrodes (or at least the one where water is consumed) are flooded and therefore without a constant gas access. Instead, hydrogen and oxygen are emitted in bubbles, which may actually be generated elsewhere from dissolved hydrogen or oxygen.

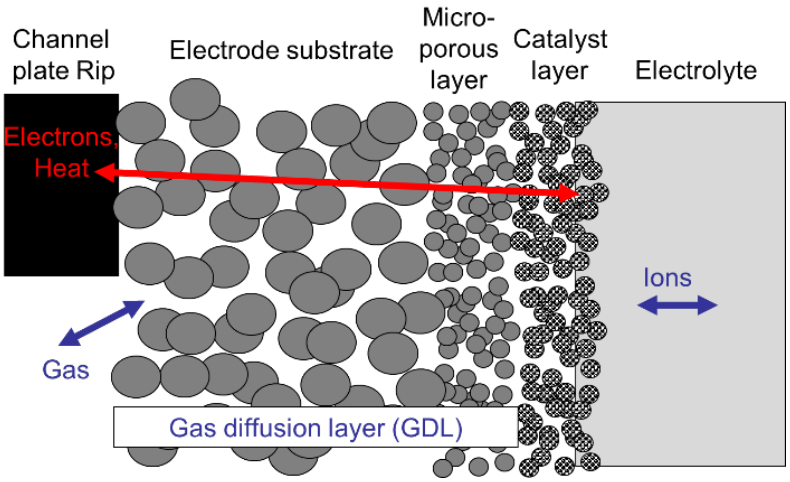


Figure 8. Principal sketch illustrating the various transport processes in electrodes. In fuel cells the porous structure outside the catalyst layer is called the gas diffusion later. In electrolyzers, this may be flooded and is called the porous transport layer.

Stacking the cells

Conversion cells are always applied in a stack of cells connected in series. The 0.6-0.8 V from single fuel cell and the 1-6-2.0 V form for single electrolyzer cells are not practical but when stacked, the voltage of all the individual cells is summed up. The current is the same throughout the stack. Another advantage of stacking is that the power density increases leading to a smaller footprint.

Many different stacking concepts have been developed, but the prevailing one is with planar cells separated by *bipolar plates* as shown in Figure 9. The term "bipolar" refers to the fact that the connect the positive electrode of one cell with the negative electrode of the neighbouring cell. They conduct electrons between the cells and jointly through the entire stack, and they secure supply and removal of reactants and products. The stack ends with a rigid end plate in each end to ensure even compression by long spring-loaded bolts or similar means. Examples of stacks are shown in Figure 10 and Figure 11.

[6] In some cases, materials with mixed ionic and electronic conductance can be applied. In other cases, a thin film of electrolyte may cover the catalyst to conduct ions. The gasses dissolve in this film and reach the reaction site as dissolved gas. In such cases, only two phases meet.

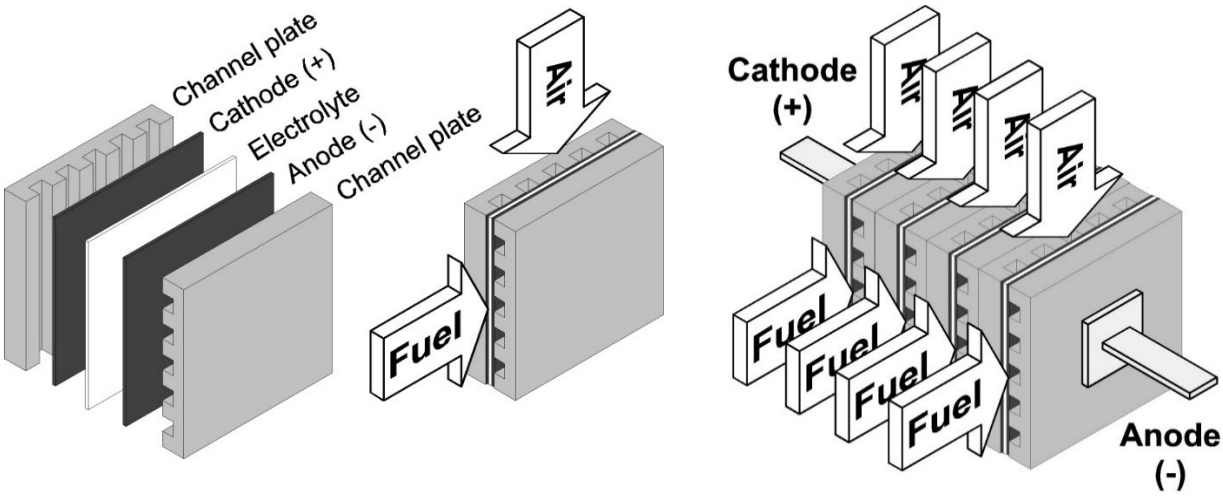


Figure 9. Stacking fuel cells. The stack is closed at each end by a rigid end plate (not shown) and fixed with long bolts. The gas channels are connected by manifolds. In fuel cells the channel design is crucial to ensure even gas distribution over the electrode. In liquid water-fed electrolyzer the design is normally different due to a focus on bubble removal.

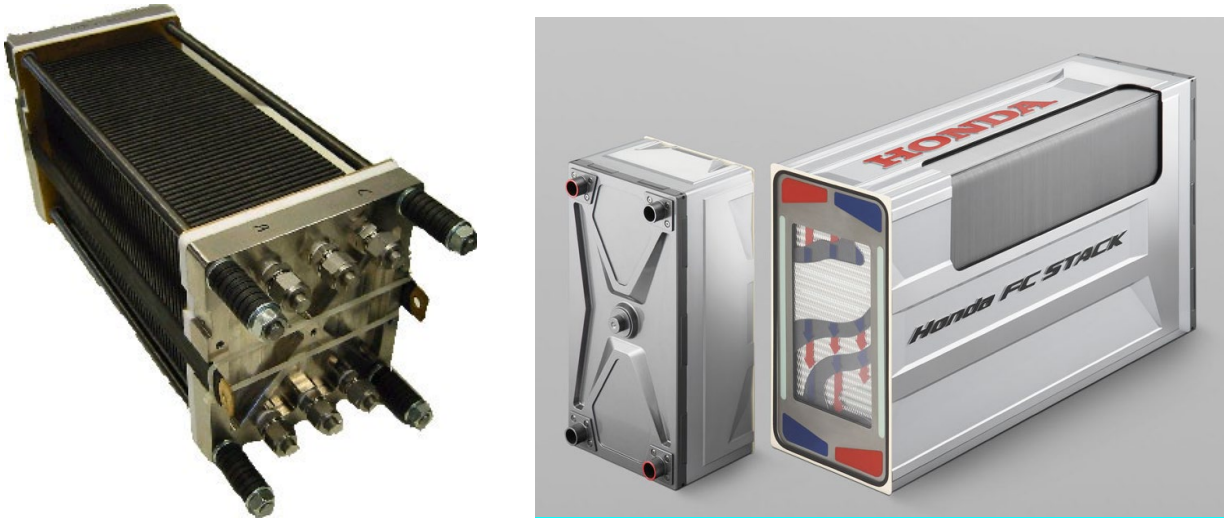


Figure 10. Fuel cell stacks. Left: Early PEM fuel cell stack from IRD fuel cells. Right: Moderns PEM fuel cell stack from Honda.

In Figure 9, the channels for inlet and outlet of gasses are open and proper gas manifolds must be applied. The usual way is to integrate gas manifolds in the bipolar plates as internal manifolds connected via channel along the stack formed when stacking the bipolar plates. See Figure 10. The long supply channels are then connected to the individual cells. The channels across the bipolar plate may take many different shapes optimized with respect to pressure drop and gas distribution. The extremes are either one long serpentine channel or straight parallel channels (like in Figure 9). In Figure 10 the pattern on the plate in font is a combination, parallel/serpentine, meant for a gas and the plate behind

has straight parallel channels for a coolant (water). Between all the cells and plates proper sealing must be applied. These can be made from elastomers (“rubbers”) for low temperature stacks and from soft glasses for high temperature stacks.



Figure 11. Electrolyzer stacks. Left, Alkaline stack from Norsk Hydro (today Nel Hydrogen). Right a PEM stack from Giner (today Plug Power).

The fuel cell system

The stack cannot work without a system around it. Flows of fuel, air and coolant are controlled by the *balance of plant* (BOP), sometimes called *balance of stack* (BOS). It is the system around the stack. A simplified BOP is shown in

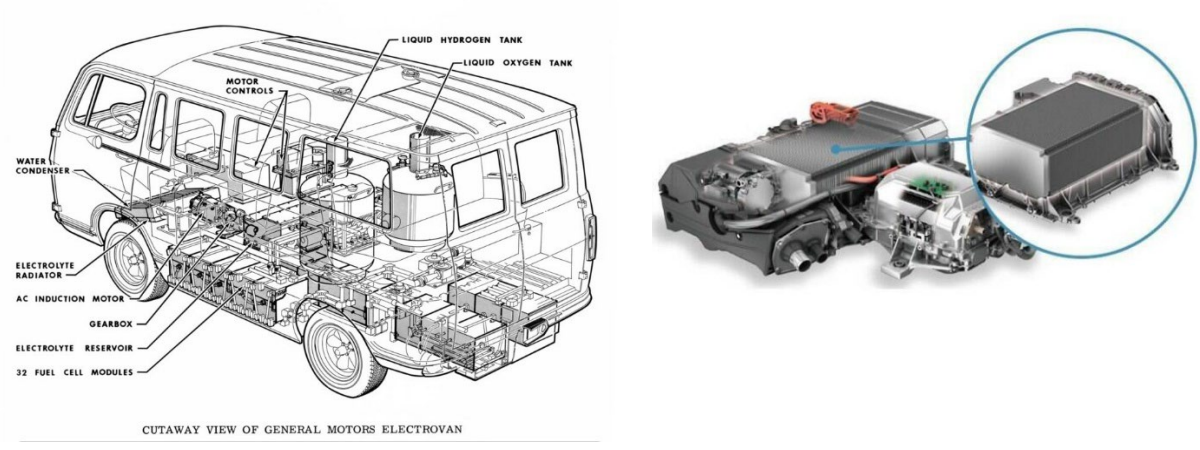


Figure 13. It can be much more complicated than in the figure, but the most important functions are shown. The electrical circuit is only shown as black box (actually orange). Depending on the application, it may consist of different units of power electronics and control devices. DC/DC and DC/AC converters

are common and normally a small battery is present as well to aid startup, stabilize the system, work as power booster and if relevant, for breaking energy recuperation.

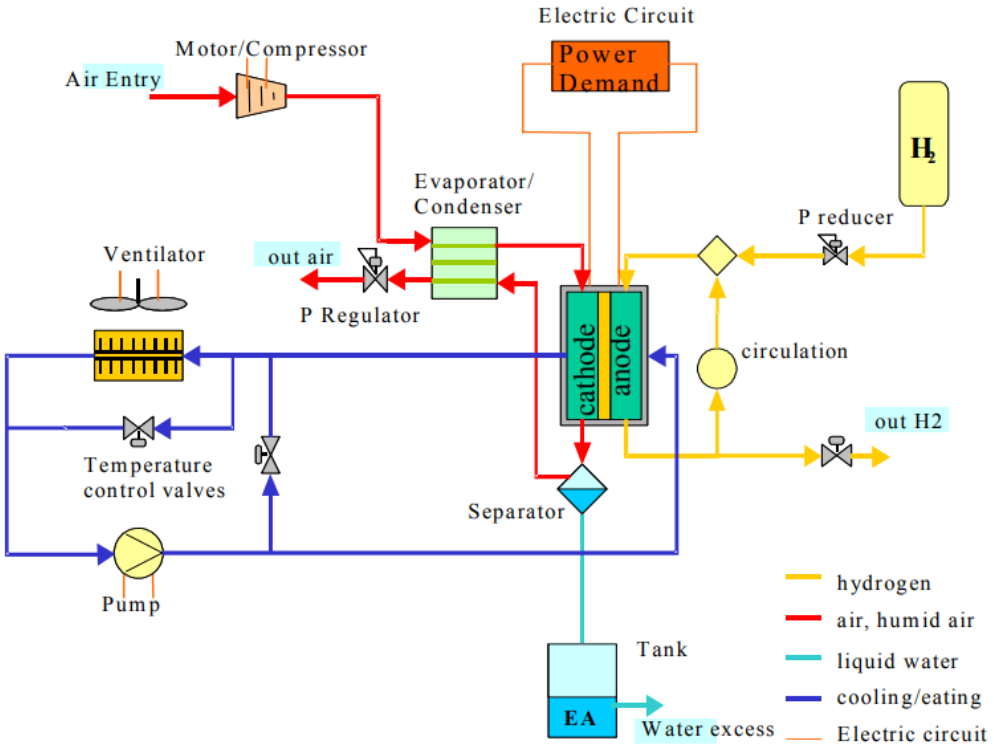


Figure 12. A typical PEM fuel cell system. Power electronics is simplified to a black box here but can be very complex.

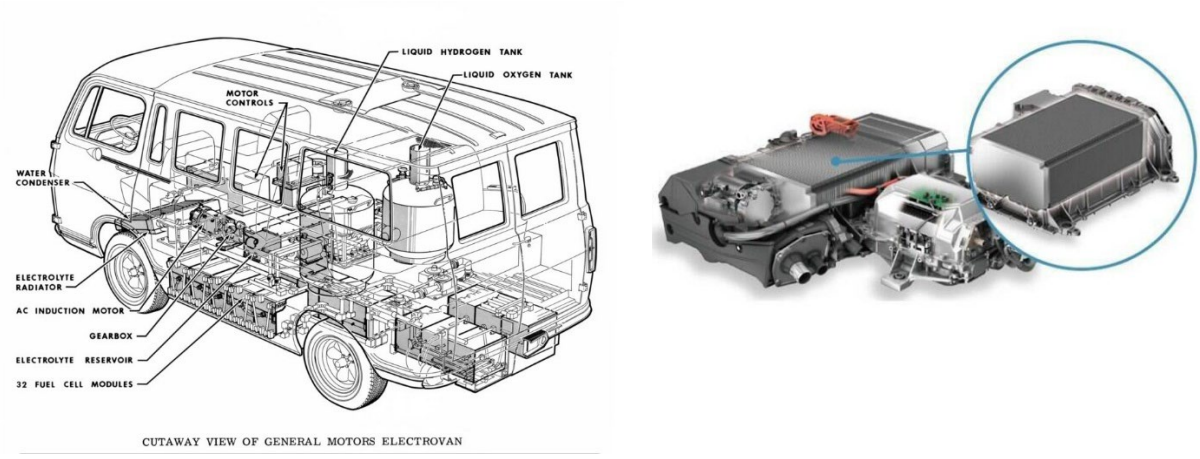


Figure 13. The fuel cell systems. Left, the system inside the first fuel cell vehicle, the Electrovan, from General Motors. Right the fuel cell system of a recent version of the Toyota Mirai fuel cell vehicle. The stack is in the emphasized circle.

The electrolyzer system

The electrolyzer system is in principle as simple as in Figure 14 but involves many additional balance-of-plant components as shown in Figure 15. The Power electronics supplying DC currents of several thousand amperes is also complicated. In addition, a water purification unit is required. If water is not very pure, impurities will accumulate in the system as water is consumed and replenished.

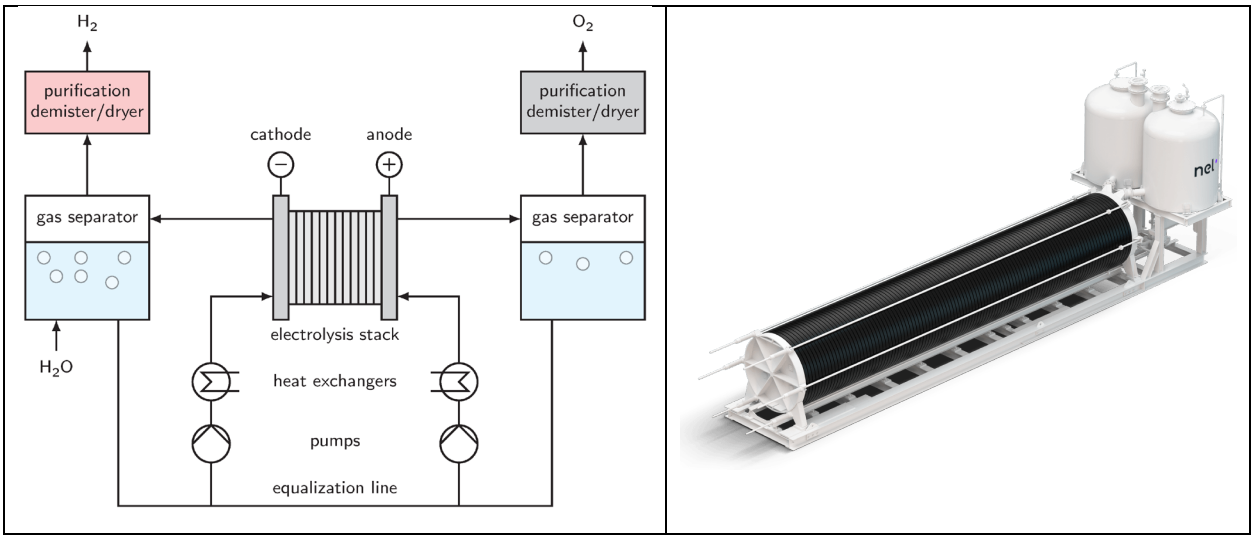


Figure 14. Left, a simplified alkaline electrolyzer system with gas liquid separator tanks. The separator tanks are clearly visible over the alkaline stack from Nel Hydrogen, left. *Sketch From Brauns, https://doi.org/10.3390/pr8020248.*

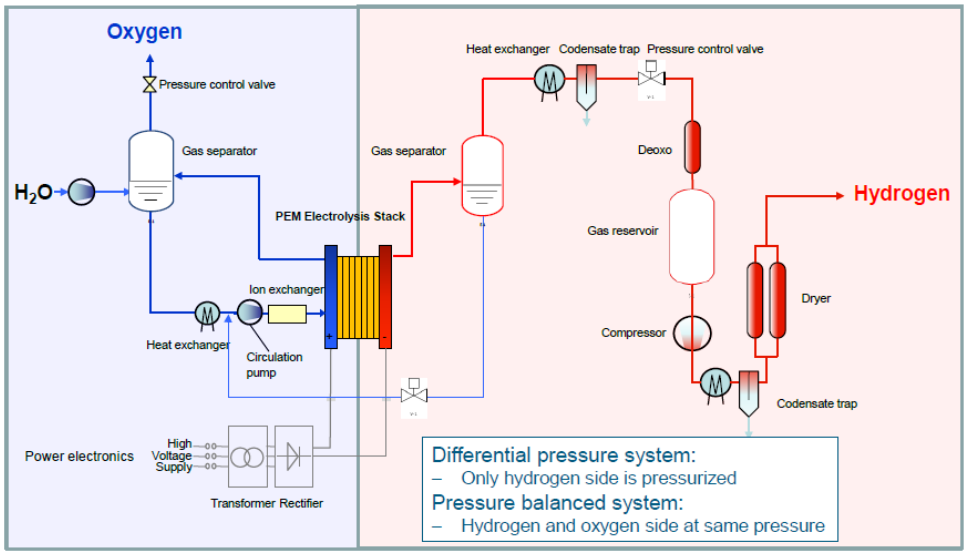


Figure 15. A PEM electrolyzer system. Alkaline electrolyzer systems are quite similar.

Thermodynamics fuel cells and electrolyzers

The relevant standard thermodynamic functions of the overall processes of a hydrogen powered fuel cell and a water electrolyzer are listed in Table 4. The total energy release or consumed in the cells is quantified by the standard enthalpy of reaction, ΔH_r^0, and the maximum electrical work obtainable or the minimum electrical work needed is quantified by the standard Gibbs free energy of reaction, ΔG_r^0.

The electrical efficiency, η, of the processes is (as the efficiency of any process) is

$$\eta = \frac{\text{What you get}}{\text{What you give}} \tag{3}$$

In the fuel cell, you get the electrical energy and give the energy of the fuel. In the electrolyzer it is opposite. Consequently, the maximum electrical efficiency of the fuel cell at standard conditions and 25 °C is given by (4)

$$\eta_{\text{max,fuel cell}} = \frac{\Delta G_r^0}{\Delta H_r^0} \tag{4}$$

And for the electrolyzer by (5)

$$\eta_{\text{max,electrolyzer}} = \frac{\Delta H_r^0}{\Delta G_r^0} \tag{5}$$

If the cells are operated at another temperature, ΔG_r^0 at that temperature must be found and used. Then (4) and (5) modifies to

$$\eta_{\text{max,fuel cell}} = \frac{\Delta G_r^0(T)}{\Delta H_r^0} \tag{6}$$

and

$$\eta_{\text{max,electrolyzer}} = \frac{\Delta H_r^0}{\Delta G_r^0(T)} \tag{7}$$

Note that the value of ΔH_r^0, is kept as that of 25 °C because the reference energy value assigned to hydrogen cannot be a function of temperature. Likewise, the energy content stated for a barrel of oil shall not depend on how anyone chooses to use it. The energy functions dependence of temperature is plotted in Figure 16 along with the reversible cell voltage which is

$$E_{\text{cell}} = \frac{\pm \Delta G_r^0(T)}{nF} \tag{8}$$

where n is the number of electrons transferred in the process (here 2) and F is Faraday's constant, the charge of one mole elemental charges (e.g., electrons). F = 96485 C mol^{-1}. The signing (±) follows that of

ΔG_r^0, i.e., "+" for the electrolyzer reaction and "-" for the fuel cell reaction. This, to get the same *positive* voltage in both cases.

The declining profile of the free energy means that the ultimate electrical efficiency (the thermodynamic efficiency) of a fuel cell also declines with temperature (eq. (6)) while the electrical efficiency of an electrolyzer increase (eq. (7)).

Table 4. Thermodynamic functions, voltages and voltage efficiency for hydrogen powered fuel cells and water electrolyzers (all species at unit activity and 25 °C). The values refer to higher heating value (HHV) or lower heating value (LHV). *Thermodynamic functions from the JANAF Series*.

	Fuel cell $H_2 + ½O_2 \rightarrow H_2O$		**Electrolyzer** $H_2O \rightarrow H_2 + ½O_2$	
	Water as liquid (HHV)	Water as vapour (LHV)	Water as liquid (HHV)	Water as vapour (LHV)
Standard enthalpy of reaction, ΔH_r^0.	-285.8 kJ mol^{-1}	-241.8 kJ mol^{-1}	285.8 kJ mol^{-1}	241.8 kJ mol^{-1}
Standard free energy of reaction, ΔG_r^0	-237.1 kJ mol^{-1}	-228.6 kJ mol^{-1}	237.1 kJ mol^{-1}	228.6 kJ mol^{-1}
Maximum voltage efficiency, η_{max}	$(\Delta G_r^0/\Delta H_r^0)$ 83%	$(\Delta G_r^0/\Delta H_r^0)$ 98%	$(\Delta H_r^0/\Delta G_r^0)$ 121%	$(\Delta H_r^0/\Delta G_r^0)$ 106%
Reversible voltage, E_{rev}^0	1.23 V	1.18 V	1.23 V	1.18 V
Thermoneutral voltage, E_{TN}^0	1.48 V	1.25 V	1.48 V	1.25 V
Minimum total energy for producing H_2 (Nm^3 at 1 atm, 0 °C)			39.38 kWh kg^{-1} 3.54 kWh Nm^{-3}	33.32 kWh kg^{-1} 3.00 kWh Nm^{-3}

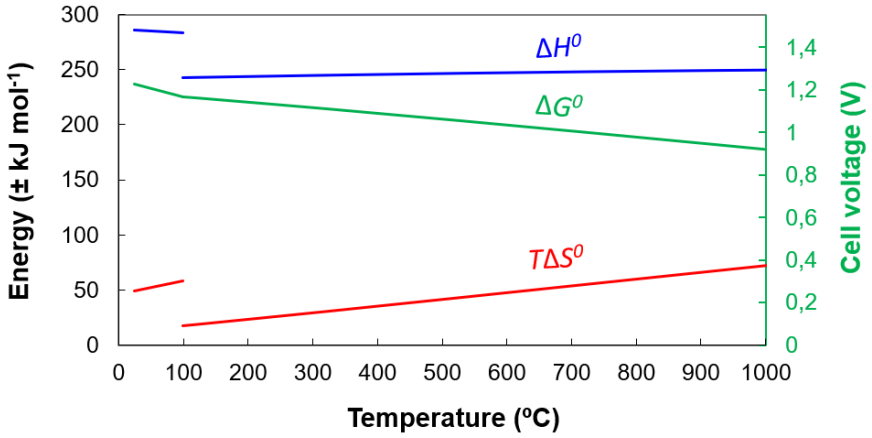


Figure 16. The energy functions of the hydrogen process (HHV). Energies are positive in electrolyzer mode and negative in fuel cell mode. On the secondary axis, the reversible cell voltage can be read from the curve of $\Delta G_{\mathrm{r}}^{0}$.

The thermoneutral voltage

We see from Table 4 that $\Delta G_{\mathrm{r}}^{0}$, the work or electrical energy, is less than the total energy, $\Delta H_{\mathrm{r}}^{0}$ of the process. This makes the fuel cell less efficient than 100%.[7] the electrolyzer on the other hand can more efficient than 100% in regard to electrical efficiency. The missing energy in the electrolyzer must be supplied as heat. The maximum heat contribution to the electrolyzer process given by the $T\Delta S^{0}$ line in Figure 16 (temperature times minimum entropy). The heat can in principle be supplied from any source, but it normally happens via additional electrical energy. Due to kinetic and ohmic imperfections in the cell, the cell voltage is always higher than the equilibrium voltage and the energy related to that is heat which is then funnelled back into the process and not wasted lie in the fuel cell.

If all the energy needed in the electrolyzer is supplied electrically, the cell voltage must be related to the enthalpy of the reaction, ΔH_{r} instead of the minimum electrical energy $\Delta G_{\mathrm{r}}^{0}$. The cell voltage corresponding to the total energy, the enthalpy of reaction, $\Delta H_{\mathrm{r}}^{0}$, can be calculates like in (8) and becomes

$$E_{\mathrm{TN}} = \frac{\Delta H_{\mathrm{r}}^{0}}{nF} \tag{9}$$

The obtained voltage is the *thermoneutral voltage*, E_{TN}. As the name indicates, a cell operated at this voltage is in thermal balance with its surroundings with no need for energy supplied as heat and no waste heat emitted. The electrical energy consumed matches exactly the chemical fuel energy of the produced hydrogen and the electrical efficiency is 100%. The thermoneutral voltage comes in two version depending on whether water is supplied as liquid or vapour. See Table 4.

[7] We know from thermodynamics that this is because operating the fuel cell results in a net reduction in the number of gas molecules and that the following loss of entropy must be balanced by dissipation of some of the energy as low temperature heat. At higher temperatures, the entropy generated by dissipation of heat is less, because the generated entropy is the heat dissipated divided by the temperature at which it happens (cf. Clausius' inequality). Consequently, the free energy of the process declines with increasing temperature (Figure 11).

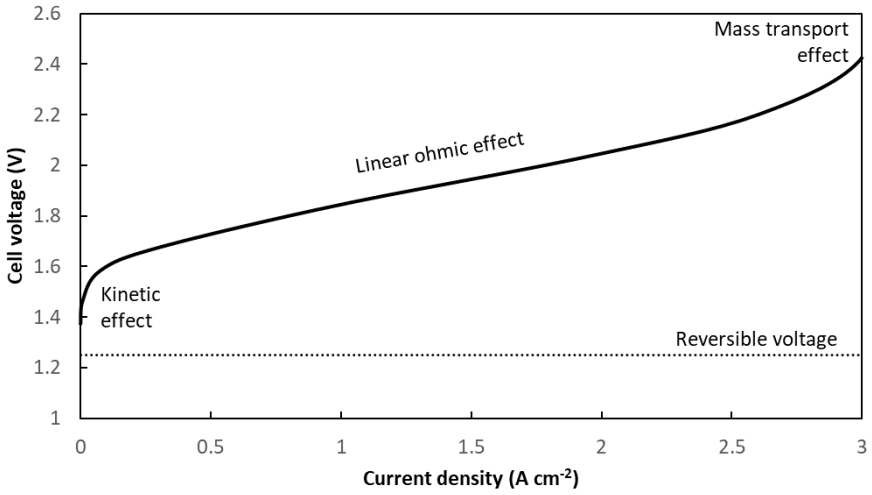
The same thermoneutral voltage applies to fuel cells even though they cannot reach this voltage. Yet is can be used as a hypothetical reference voltage corresponding to 100% electrical efficiency.

Performance and actual efficiency

Polarization curves

Fuel cells and electrolyzer cells are characterized by their polarization curves or IV curves (after current and voltage). Voltage is plotted as a function of the current – or commonly the current density which is the current normalized to the area (e.g., amperes per cm^2). Examples of polarization curves are shown in Figure 17. Such curved can be quite different numerically, but they follow the same shape.

The polarization curve begins at zero current density close to the reversible voltage by facing an immediate significant voltage shift away from the reversible voltage. This is due to the electrode processes, mostly that for oxygen reduction and can be labelled a kinetic overvoltage. In the successive linear part, the kinetic drop does not develop further, and the curve is dominated by a purely ohmic development mostly caused by the electrical resistance in the electrolyte (ohm's law). If the curve begins to bend further at high current density, it indicates problems of mass transport, i.e., transport of reactants to the reaction sites. If this transport cannot keep up with the consumption, the cell voltage cannot be maintained its trend and shifts further away from the reversible voltage.



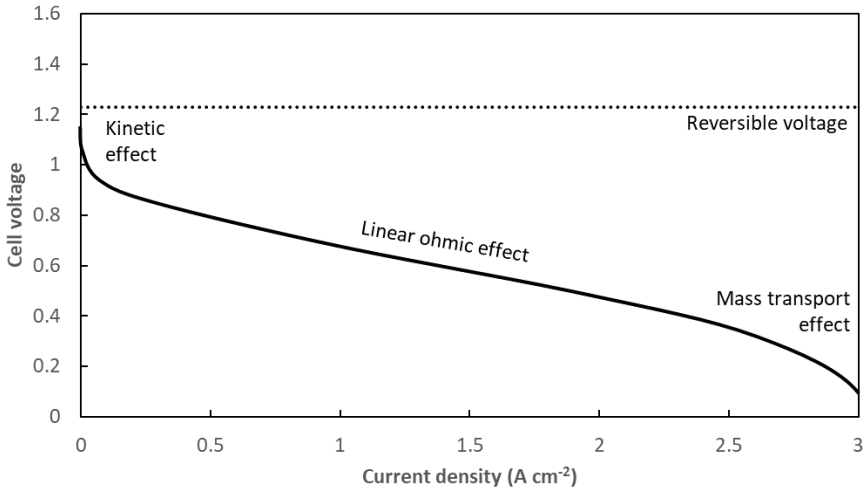


Figure 17. Polarization curves for an arbitrary electrolyzer (top) and a fuel cell (bottom).

Actual efficiency

So far, we have only treated the idealized thermodynamic efficiency of the conversion cells. The actual efficiency of an operating cell or of the system is what matters practically. Different efficiencies, which each tells part of the story, are defines as in Table 5 and elaborated below.

Table 5. The most important efficiencies of fuel cells and electrolyzers.

Efficiency	**Description**	**Applies to**
Thermodynamic efficiency	The thermodynamic limit for the electrical efficiency	Fuel cells, eq. (6) Electrolyzer cells, eq. (7)
Faradaic efficiency (current efficiency)	The degree of fulfilment of Faraday's law.	Fuel cells: The efficiency of conversion of the fuel molecules to electrons. Electrolyzers: The efficiency of conversion of electrons to product molecules
Voltage efficiency	The ration between the actual cell voltage and the idealized voltage determined by thermodynamics.	Fuel cells and electrolyzer cells.
Fuel efficiency	The fraction of the fuel that is consumed in a fuel cell.	Fuel cells only. In some cases, a small fraction of the fuel is passing unconverted.
System efficiency	The total electrical efficiency of the entire system including the balance of plant	Fuel cell systems and electrolyzer systems.

Thermodynamic efficiency is already treated and is quantified in the equations (6) and (7).

Faradaic efficiency is the current efficiency. Evet pair of electrons should correspond to one hydrogen molecule, and this holds largely for both fuel cells and electrolyzers. This relationship is expressed by Faraday's law (10) where *I* is the current, *n* is the number of mole electrons in the reaction, and $\dot{n}_{\mathrm{H2}}$ is the molar flow of hydrogen.

$$\text{Faraday's law: } \dot{n}_{\mathrm{H2}} = \frac{I}{nF} \tag{10}$$

(10) can also be written with current density and then the molar flow is flow per cell area.

The Faradaic efficiency is the ration between the actual production of hydrogen or current and the theoretical production of the same according to (10).

For a **fuel cell**, the faradaic efficiency, $\eta_{\mathrm{F,FC}}$, is the ratio between the actual current and the ideal current given by the fuel consumed and eq. (10).

$$\eta_{\mathrm{F,FC}} = \frac{I_{\mathrm{actual}}}{I_{\mathrm{ideal}}} \tag{11}$$

This can be combined with (10) to give

$$\eta_{\mathrm{F,FC}} = \frac{I_{\mathrm{actual}}}{nF\dot{n}_{\mathrm{H2}}} \tag{12}$$

For an **electrolyzer cell,** it is (as always) the other way round and the faradaic efficiency, $\eta_{\mathrm{F,EC}}$, becomes

$$\eta_{\mathrm{F,EC}} = \frac{I_{\mathrm{ideal}}}{I_{\mathrm{actual}}} \tag{13}$$

and

$$\eta_{\mathrm{F,EC}} = \frac{nF\dot{n}_{\mathrm{H2}}}{I_{\mathrm{actual}}} \tag{14}$$

Again, current can be replaced with current densities if the hydrogen flow is area specific.

The **voltage efficiency** is the most critical one in most cases. It relates to the overvoltages of the electrokinetic electrode processes and the ohmic voltage loss across the electrolyte. Quantification of that is most easily done with a reference voltage corresponding to 100% electrical efficiency and this is the thermoneutral voltage, E_{TN}, eq. (9). The voltage efficiency of fuel cells and electrolyzer cells, $\eta_{V,FC}$ and $\eta_{V,EC}$, respectively, become

$$\eta_{V,FC} = \frac{E_{actual}}{E_{TN}} \tag{15}$$

and

$$\eta_{V,EC} = \frac{E_{TN}}{E_{actual}} \tag{16}$$

Voltage efficiency can be determined form the polarization curves. This is illustrated in Figure 18 and Figure 19.

For fuel cells, the voltage efficiency is normally moderate due to overvoltages, say 40-60%. For electrolyzers the voltage efficiency is higher despite electrolyzers have similar overvoltages. This is because the reference voltage is the thermoneutral voltage, which is higher than the reversible cell voltage so, higher voltage losses can be accommodated before the voltage efficiency drops significantly below 100%. Typical voltage efficiencies of electrolyzers are 70-85%, and high-temperature electrolyzers show practical voltage efficiencies around 100%.

Fuel efficiency is only relevant to fuel cells. If hydrogen is mixed with other gasses, the cell cannot use all, and some with pass unused through the cell.

System efficiency is the energy efficiency of the entire system including balance of plant (the system around the electrochemical cells). End of the day this is the key performance indicator to the user. For electrolyzer systems, this is often stated as the electrical energy consumption for the production if one normal cubic metre of hydrogen (Wh Nm^{-3}). The numbers are listed in Table 4. As always there are different numbers depending on whether water is supplied as liquid or vapour, but the user need not think about that. If it is stated that an electrolyzer system consumes 4.5 kWh Nm^{-3}, this will correspond to 3.54/4.50 = 0.79 = 79% (HHV) or 3.00/4.50 =0,67 = 67% (LHV).

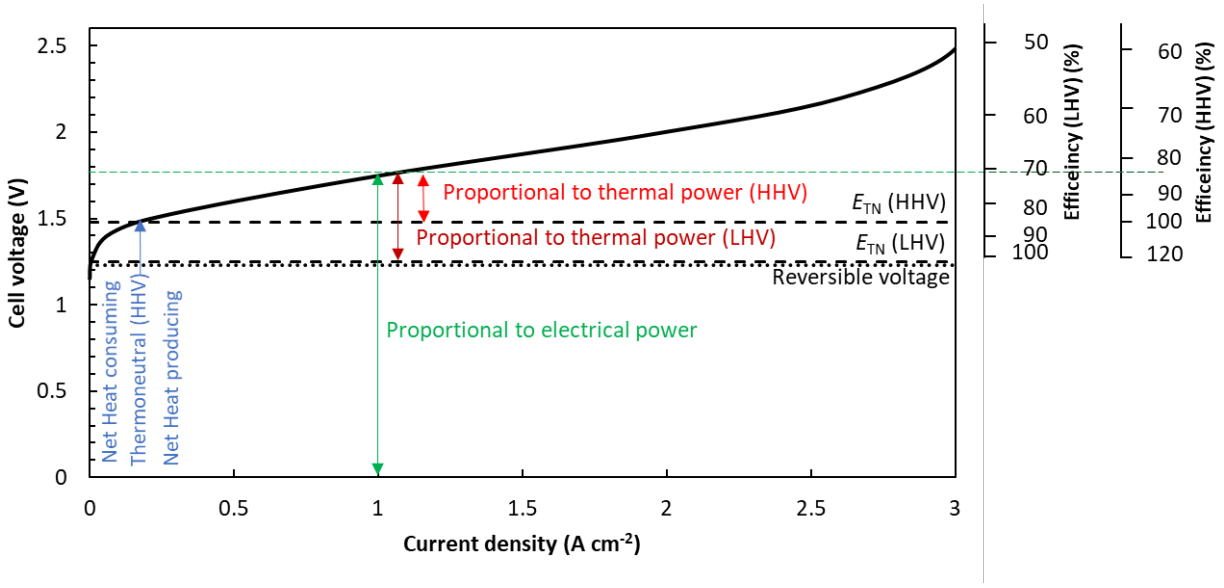


Figure 18. Reading voltage efficiency of an arbitrary fuel cell from its polarization curve.

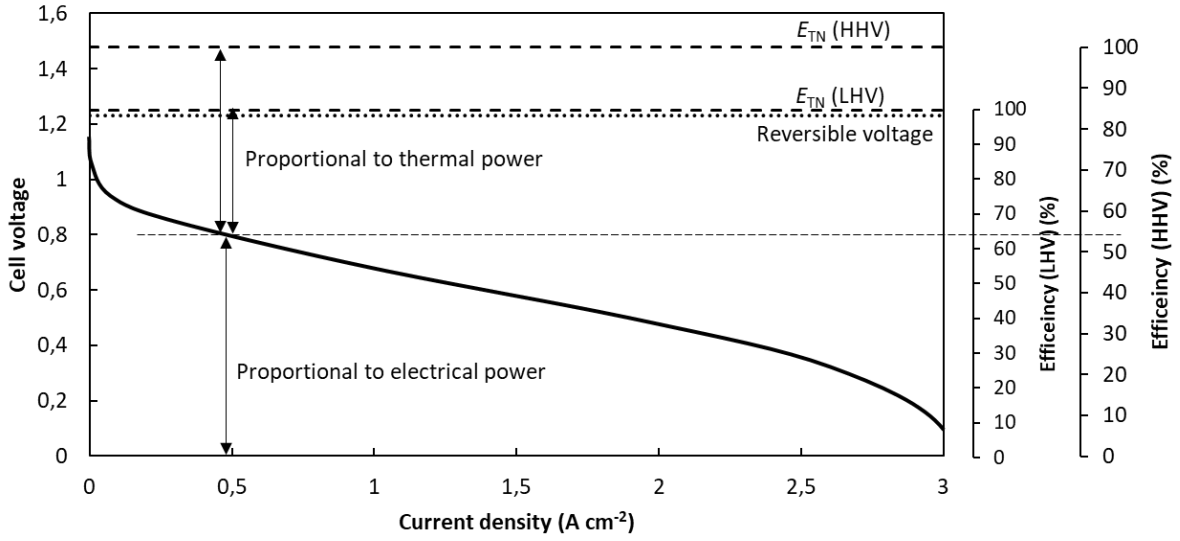


Figure 19. Reading voltage efficiency of an arbitrary fuel cell from its polarization curve.

Synthetic fuels

The term "*synthetic fuel*" or just "*synfuel*" means fuel that is made synthetically as compared to fuels extracted from natural reserves e.g., in oil and gas fields or coal mines. It can be questioned if cracked and refined petrochemical products are less synthetic, but the term synthetic fuel is well established and used for the variety of fuels that can be built from simple starting materials, which can be hydrogen and CO or CO_2. Today, the hydrogen is often expected to stem from electrolysis driven by renewable electricity generated from wind or solar, but hydrogen from gasification of coal is also considered. Then it is not part of the green transition, but just a possible business case or a necessity in an isolated economy with limited oil reserves. Examples of the latter are Germany during World War 2 where fuel for the war machine was synthesized from coal. A similar solution was adopted in South Africa during the apartheid embargo. Synfuels in which the energy comes from electricity are also termed "*electro-fuels*" and at time even "*liquid electricity*", perhaps with a smile. Hydrogen formally falls within the definition of a synthetic fuel, but in many contexts, it is carbonaceous fuels that are referred to.

When hydrogen is processed further to methane, methanol or gasoline by reaction with CO or CO_2, some energy is lost in the synthesis process, and the final fuel contains a little less energy than the original hydrogen. This can make it look silly to make the effort, and strictly energetically, it doesn't make sense. However, the benefit is a fuel that is easier to store, transport and fill. The energy density is higher than of hydrogen regardless of the way hydrogen is stored. This is important for energy intense mobile applications like heavy transport and aviation. Moreover, and in contrast to hydrogen, many synfuels fit into the existing fuel infrastructure, perhaps with minor modifications. These advantages might be decisive, but for applications where hydrogen can be used, this is clearly the most energy efficient solution. The energy density (energy pr. volume) and the specific energy (energy pr. mass) of

some common fuels are plotted in Figure 20. Hydrogen, batteries and pumped hydro are added for comparison. Pumped hydro is water elevated to a high level, typically a mountain reservoir, storing the energy as potential energy, which can be reclaimed with high efficiency via a turbine.

Basically, what it takes to make synthetic fuels is a carbon source and hydrogen. If the carbon source is itself a fuel (coal, heavy oil or other fossil fuels that are not easily serving a given fuel purpose) then the chemical energy is already there. If the carbon source is CO_2, then the energy of the fuel that is the desired outcome of the process, must be provided by other sources, typically hydrogen and perhaps to some extend heat.

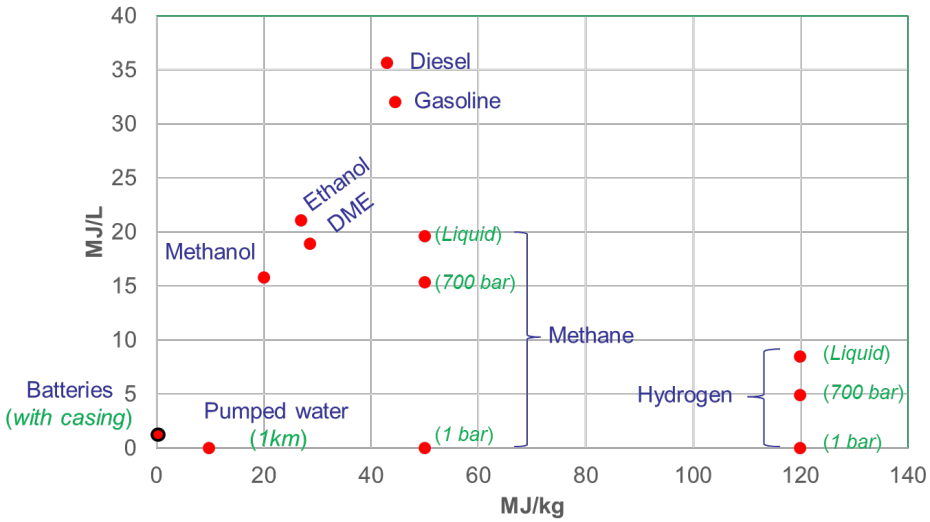


Figure 20. Energy density (energy pr. volume) and specific energy (energy pr. mass) of selected fuels.

The Sabatier process for methane

The Sabatier process (named after *Paul Sabatier*) is a process for the synthesis of methane from hydrogen and carbon dioxide

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \qquad \Delta H_r = -165.0 \text{ kJ mol}^{-1} \tag{17}$$

It is carried out at 300-400 °C over a nickel catalyst. The product is often referred to as "*synthetic natural gas*" (SNG) although it is just methane. One obvious use of the process is for the methanation of the CO_2 of biogas. Biogas is made by fermentation of organic substances like farming waste of all kinds. The microorganisms consume some of the chemical energy and produce CO_2. Additional CO_2 is formed by the oxygen already present in starch, cellulose and other biological substances. Typically, around 50% of biogas is CO_2, so upgrading to CH_4 represents a significant boost of the energy content. The CO_2 is moreover a high concentration source of CO_2 for synthetic fuels. A nice feature of synthetic natural gas is that it can be handled by the natural gas infrastructure that is already present in many countries, especially natural gas grids and pipelines.

Fischer-Tropsch synthesis for longer hydrocarbons

When it comes to larger hydrocarbons for liquid fuels like synthetic gasoline and jet fuel, the way is to build the chains up carbon by carbon. The technology was originally developed by *Franz Fischer* and *Hans Tropsch* in 1925 and is named after the inventors. The building blocks are then CO and H_2. A mixture of CO and H_2 is called *syngas* (synthesis gas) and by tuning the ratio of the two components and the process condition and with a proper choice of catalyst, a variety of substance can be synthesized. If the carbon source is CO_2, it is first converted to CO by the *reverse water gas shift reaction* or by electrolytic reduction of CO_2

$$CO_2 + H_2 \rightarrow CO + H_2O \qquad \Delta H_r = +41 \text{ kJ mol}^{-1} \qquad (18)$$

Then follows the Fischer–Tropsch process:

$$(2n + 1)H_2 + nCO \rightarrow C_nH_{2n+2} + nH_2O \qquad \Delta H_r << 0 \qquad (19)$$

The product is a mix of many different hydrocarbons, and it might be necessary to crack the largest molecules into the right size for the desired fuel. This was the technology used in Germany during World War II and in South Africa during the international embargo in the apartheid days, and today South Africa has a significant production of fuels via Fischer-Tropsch.

Methanol

Methanol is another fuel (or chemical) that can be produced from syngas. The reaction takes place over a copper-based catalyst at 200-220 ^oC and moderate pressure after

$$CO + 2H_2 \rightarrow CH_3OH \qquad \Delta H_r = -90.64 \text{ kJ mol}^{-1} \qquad (20)$$

or

$$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O \qquad \Delta H_r = -49.47 \text{ kJ mol}^{-1} \qquad (21)$$

Ammonia

Ammonia production is the largest application of hydrogen today and most of the ammonia is used as the nitrogen source in synthetic fertilizers. However, ammonia can also be regarded as a fuel. Ammonia is a gas at standard conditions, but the vapour pressure is only around 10 bar at room temperature so at that pressure it is liquefied and easily stored as such.

Ammonia is produced in large plans by the Haber-Bosch process (after *Fritz Haber* and *Carl Bosch*) which is a direct reaction between hydrogen and nitrogen after

$$3H_2 + N_2 \rightarrow 2NH_3 \qquad \Delta H_r = -92.4 \text{ kJ mol}^{-1} \qquad (22)$$

at 100 – 250 bar and 400 - 500 °C over an iron-based catalyst. Since the number of gas molecules decline in the reaction, high pressure favours the product. High temperature favours the reaction rate, but it also shifts the reaction equilibrium towards the reactants (negative reaction enthalpy). Consequently, temperature is a compromise between reaction rate and equilibrium.

The obvious disadvantage of ammonia is the strong and unpleasant smell and the fact that it is harmful and even lethal at certain concentrations in the air. The very attractive feature is that it does not contain carbon. Thus, it doesn't emit CO_2 and extraction of N_2 from air is so much easier than of CO_2 because four out of five molecules in the air are N_2. If ammonia is to be produced from renewable hydrogen, there is no dependence on point source with CO_2 at practical concentrations, like cement factories and traditional coal fired power plants.

Ammonia can be combusted as is with oxygen after

$$4NH_3 + 3O_2 \rightarrow 6H_2O + 2N_2 \quad (23)$$

It is also easily converted directly in a solid oxide fuel cell, but before use in a PEM fuel cell it must be converted back to hydrogen by cracking, which is the reverse of the formation process above. This requires a temperature of at least 300 °C both for kinetic reasons and for the equilibrium composition to be in favour of H_2 and N_2. Ammonia is considered as a future fuel for certain large scale uses, like in ships. Maritime internal combustion engines do not work as easily on ammonia as on carbonaceous fuels, but the technology is being developed. For commuter vehicles and smaller scale applications it is questionable whether the safety issues can be handled satisfactory.

Carbon capture

Carbonaceous synfuels require a carbon source, and CO_2 which would otherwise be emitted, is normally considered carbon neutral. This is based on the simple assumption that it just makes another fuel cycle before it is emitted. This makes sense, but energy used for collecting the CO_2, for the handling and for the synthesis processes should be carbon neural as well for the assumption to hold completely.

Carbon capture is easiest and least energy consuming when done from point sources. These are sources with high large amounts at high concentration or high partial pressure. Common examples are cement factories, biogas facilities and conventional power plants. If such sources are not fully utilised, they are the first choice. On the longer term, they will not suffice and capturing from the ambient air may be necessary. This is called *direct air capture* (DAC) although it is not the air that is captured.

In the literature, distinction is made between *pre-combustion* and *post combustion* capture, but the caption process is in principle the same. Pre-combustion relates to CO_2 from a combustion process, but in a setup where CO_2 is separated after a reforming process where the fuel reacts with water, forming hydrogen and CO_2 prior to the combustion process (combustion of the formed hydrogen). Then CO_2 is not diluted with nitrogen as it is after combustion with air. Another process leading to high CO_2

concentration is the *oxy-fuel process* in which the fuel is combusted with pure oxygen. The oxygen can be the product of electrolysis.

Capturing can be performed by *absorption*, *adsorption* or *filtration*. The difference between absorption and adsorption is that adsorption occurs at a surface while absorption occurs in the bulk of a liquid, a solution or a solid. A common term for both processes is *sorption*. Separation via sorption can be *pressure swing* or *temperature swing* in which either temperature or pressure (or both) is changed to shift from sorption mode to desorption mode. In some techniques where water is involved in the sorption process, sorption and release can be driven by *moisture swing*. The Swiss company Climeworks develop direct air capture plants with large CO_2 collectors.

Absorption

The state of art CO_2 adsorbing medium is amines like monoethanolamine, $NH_2CH_2CH_2OH$ (MEA), diethanolamine, $NH(CH_2CH_2OH)_2$ (DEA), and methyldiethanolamine, $CH_3N(C_2H_3OH)_2$ (MDEA). These are liquids and CO_2 binds as shown in Figure 21. The binding process is exothermic, and CO_2 is later released by heat at higher temperature.

The lower the concentration of CO_2, the stronger bonding is needed for effective absorption. Stronger absorbents are hydroxides, e.g., $Ca(OH)_2$ and NaOH. Solutions of these inorganic salts readily absorb CO_2 from air, but the downside of the strong bonding is that it also takes more energy and higher temperature to release the absorbed CO_2.

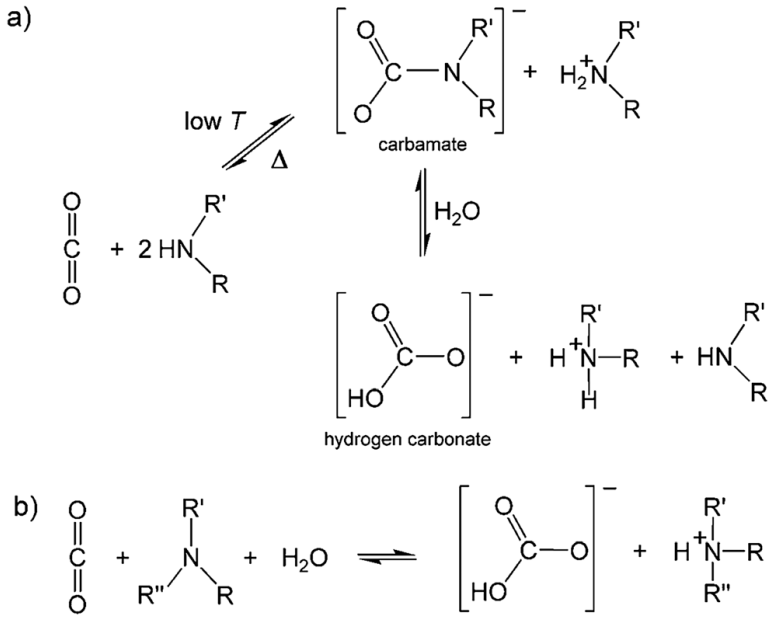


Figure 21. The binding and release of CO_2 by amines. R, R' and R'' are hydrogen, methyl groups or radicals of ethanol.

Adsorption

Adsorption requires a material with a high surface area. Common classes of high surface area materials are various carbon materials (active carbon, carbon nanotubes etc.), zeolites and metal-organic frameworks (MOF). Zeolites and MOFs are inorganic and organic/inorganic structures, respectively, with very well-defined pore sizes. See Figure 22 and Figure 23. They can be tailored to selective adsorption of CO_2. Regeneration, i.e., CO_2 release is done by pressure or temperature swing.

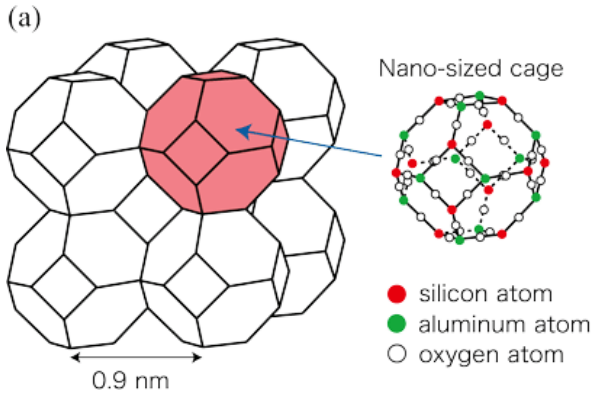


Figure 22. An example of a zeolite. Specific structures are formed by the mixed oxides of cations of different charge and size. The pore shapes and sizes are well-defined and given by the choice of positive ions and their ratio. The shown structure continues in all directions.

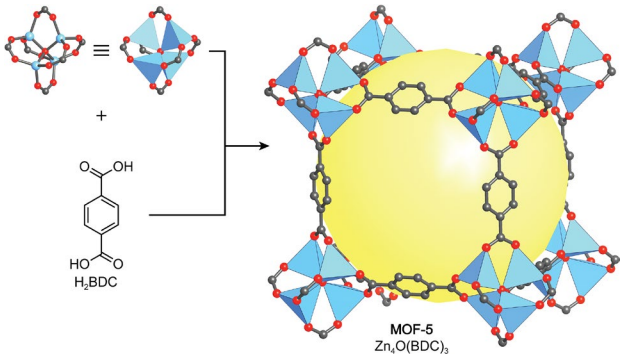


Figure 23. An example of a metal-organic framework (MOF). The blue tetrahedral arrangements are zinc ions coordinated with four oxygen ions. The inorganic corner structures (nodes) made around metal ions are connected by organic bridges. The shown structure continues in all directions.

Membrane filtration

Semi-permeable membranes can be used to separate CO_2 from a gas mixture. The selectivity is caused by a combination of pore sizes and affinity to specific molecules in the membrane. The driving force for the passage through the membrane is the partial pressure difference across. Therefore, membrane filtration of CO_2 for direct air capture requires a very high total pressure to make the partial pressure of CO_2 high enough and then the partial pressure of more abundant gasses becomes even higher, and the CO_2 selectivity must be very significant.